

CROVIA

Crowd safety from telecom network signals



REAL-TIME DETECTION

Identifies dangerous crowd build-ups using Nokia Network as Code (Open Gateway / CAMARA) APIs.



PRIVACY-FIRST TECHNOLOGY

Operates without cameras or hardware; the mobile network acts as the primary sensor.



PILOT CITY: (LUSAIL, QATAR)

30,000 people across 4 districts and 5 monitored zones.



GSMA MENA IGNITE HACKATHON PROJECT

The Arithmetic of Crowd Crushes

Why invisible systemic pressure leads to disaster



THE BOTTLENECK

At Lusail Stadium, the north concourse narrows to a 9-metre ramp, creating a severe physical constraint.



PHYSICAL LIMITS

The ramp has a maximum throughput of 648 people per minute.



THE INFLOW GAP

With an arrival rate of ~1,200/minute, there is an excess of 552 people every minute with nowhere to move.



THE DANGER THRESHOLD

The ramp reaches a dangerous capacity of 5,297 people in approximately ten minutes.

THE VISIBILITY TRAP: THIS ACCUMULATION IS INVISIBLE TO THOSE ON THE GROUND; PARTICIPANTS ONLY SEE IMMEDIATE CROWD DENSITY, NOT THE SYSTEMIC INFLOW PRESSURE.

Why Current Safety Tools Fall Short

The structural gap in existing crowd management technologies

Tool	Coverage Limit	Latency	Failure Mode
CCTV	Camera field of view	Requires human monitor	Un-instrumented areas
Drones	Flight window dependent	Post-concern launch	Unplanned crowds
Manual Management	Steward location	Human observation lag	Rate-based surges
Wi-Fi/BT Counters	Fixed sensor range	Near real-time	Devices switched off
Ticketing	Entry points only	Real-time at gate	Location/speed tracking
GPS Apps	Opt-in users only	Real-time	Indoor/Privacy refusal

EXISTING TOOLS TRACK STATIC TOTALS; CROVIA TRACKS FLOW RATES, PROVIDING INTELLIGENT ALERTS NINE MINUTES BEFORE INCIDENTS OCCUR.

One Piece of Arithmetic That Needs No Sensor

Capacity per minute = Width in Metres x 72

Zone	Bottleneck	Width (m)	Passes/Min	Holds
Stadium north concourse	North ramp narrows	9.0	648	5,297
Stadium tram link	Tram platform	7.5	540	5,850
Marina promenade	Promenade walk	5.0	360	7,243
Fox Hills crossing	Signalised crossing	8.0	576	3,874
Central plaza crossing	Plaza crossing	10.0	720	15,057

Why Open Gateway APIs Are Essential, Not Convenient

Leveraging network-native capabilities for superior performance and privacy

- **NETWORK-SIDE LOCATION INTELLIGENCE**

Enables precise positioning without handset cooperation or active GPS, bypassing limitations like indoor signal drops or high crowd density.

- **PROPRIETARY CONGESTION INSIGHTS**

Provides exclusive access to real-time, push-based cell load data that is inaccessible to third-party providers.

- **ACTIVE NETWORK CONTROL**

Utilizes 'Quality on Demand' to move beyond passive observation, allowing for prioritized connectivity during high-saturation events.

- **OPERATOR-ENFORCED CONSENT**

Centralizes permission management at the subscriber level for a more transparent and legally robust privacy framework.

THE NETWORK IS THE ONLY EXISTING, PRE-FUNDED SENSOR INFRASTRUCTURE THAT REQUIRES NO INSTALLATION AND POSSESSES THE UNIQUE ABILITY TO ACT ON THE DATA IT GATHERS.

How the Trigger Logic Works

A three-signal framework for district analysis



CONGESTION INSIGHTS

Acts as the primary trigger, providing congestion levels (None to High) while maintaining anonymity regarding device identity or location.



GEOFENCING

Applies district-level context to congestion signals to avoid the inaccuracies inherent in tight-zone tracking.



LOCATION VERIFICATION

A premium signal providing unbiased estimates via random sample-based calculations: $(\text{inside} / \text{checked}) * 30,000$.



HARD AND LOGIC

Triggers investigation if congestion share is $\geq 10\%$ and $\geq 2x$ the calmest district, or $\geq 25\%$ independently.



RELATIVE COMPARISON

Uses district-to-district benchmarks to account for inherent differences between urban and suburban baselines.



TREND ANALYSIS

Employs a least-squares slope over a seven-minute window to filter sampling noise and ensure reliable fill-rate detection.

Why a Busy Place is Not an Alarm

Understanding the criteria for active safety alerts



ACTIVE ALARM REQUIREMENTS

An alarm triggers only when six conditions are met simultaneously: 60% bottleneck capacity, two consecutive high readings, 120-second duration, 1,200+ people, 1.4+ risk factor, and 2x baseline variance.



AUTOMATIC SILENCING RULES

The system ignores 'draining flow' if cohorts move at 75% speed and suppresses synchronous network-wide spikes to prevent false positives.



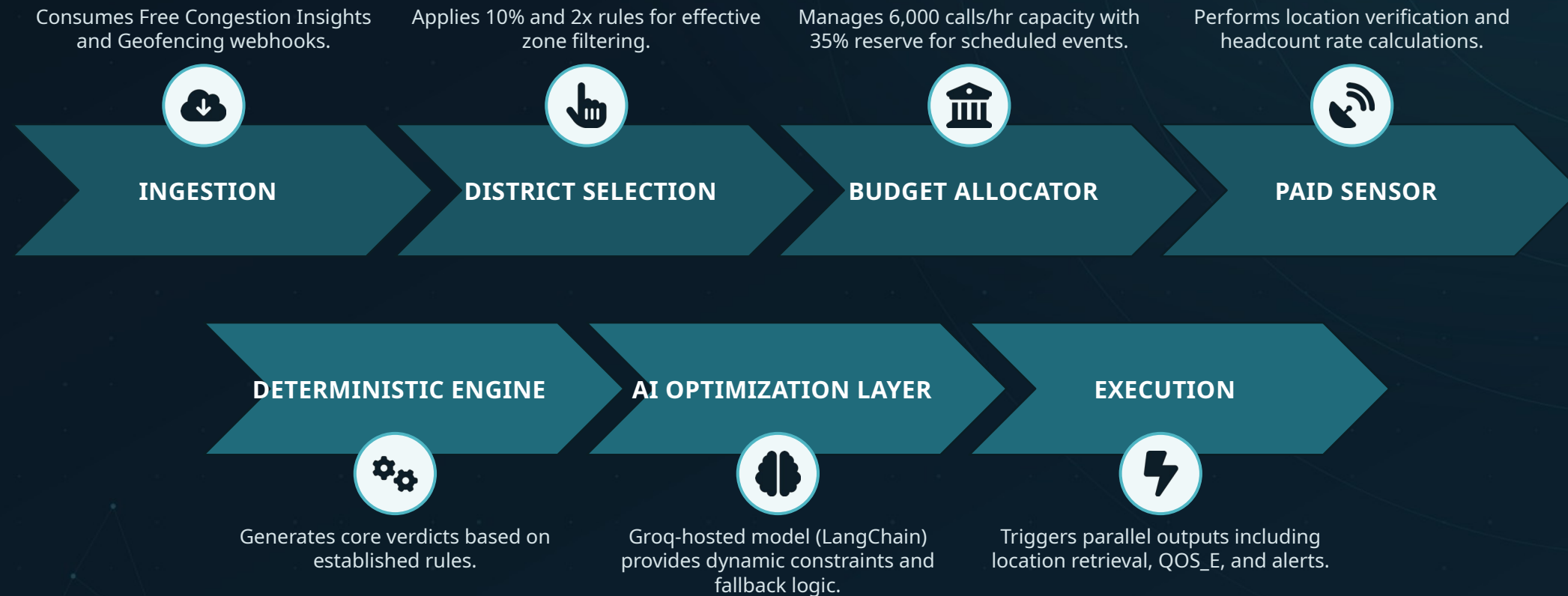
CASE STUDY: STADIUM ENVIRONMENT

In an 18,000-spectator scenario, the system remains silent despite high congestion because the near-zero arrival rates fail to reach the 60% capacity threshold.

THE SYSTEM PRIORITIZES ACTIONABLE RISK OVER MERE CONGESTION BY FILTERING OUT NORMAL FLOW AND NETWORK NOISE.

Signal Processing & AI Decision Pipeline

Integrated Architecture for Automated Resource Allocation



SYSTEM BOUNDARY: THE AI OPTIMIZES RESOURCE ALLOCATION BUT DOES NOT DETERMINE HUMAN SAFETY STATUS.

API Usage Synopsis

Optimization Strategy: Escalation Logic, Priority, and Data Integrity

API	Function	Role	Relative Cost
Congestion Insights	Push when cell is loaded	Decides where to look	1.0 (sub)
Geofencing	Push on entry, exit, sub-end	Gives address	1.0 (sub)
Location Verification	Is device inside circle?	Main sensor (headcount)	1.0
Device Status	Can device answer?	Free pre-filter	0.5
Location Retrieval	Actual coordinates	Alarm time trigger	3.0
Quality on Demand	Protect responder connection	Network modifier	5.0

OUR STRATEGY EMPLOYS A CHEAPEST-FIRST APPROACH, PRIORITIZING DATA INTEGRITY BY DISCARDING UNKNOWN RESPONSES TO AVOID UNDER-REPORTING, WHILE RESERVING HIGH-COST OPERATIONS ONLY FOR VERIFIED SURGES.

Privacy and Consent by Design

Standardized protocols for data sovereignty and safety

- **CONSENT MODEL**

Consent is captured at sign-up, in the same request as enrolment — silence is never agreement.

Enrolment is refused without an active consent record; there's no path around it.

Withdrawal stops monitoring instantly and is recorded, never deleted, for full auditability.

- **DATA HANDLING**

Real numbers are SHA-256 hashed to a 16-character ID; every system downstream uses only that hash.

The real number surfaces only at the last moment, right before a Nokia call.

Coordinates are bought once, at alarm time, for ~20 devices — never during normal monitoring.

k-anonymity of 5, a 400-device counting panel, and a 500-device locating pool mean most people are never tracked.

Only two roles exist — citizen and authority — and neither ever sees raw phone numbers.

- **ESCALATION PROCESS**

Everyone placed in the affected district is warned — an unknown location is never treated as safe.

Warnings cap at one per person, per place, per 15 minutes, so people don't learn to ignore them.

Human-in-the-loop is an operator acknowledgement after the warning goes out, never a gate before it.

Two consecutive falling readings stand an alarm down with an all-clear; position data never cancels an alarm.

Open Gateway APIs: Current Implementation vs. Future Scale

Evolution from sandbox proof-of-concept to production-grade deployment

Built Today



Current capabilities include Congestion Insights, Geofencing, Location Verification, Device Reachability, Location Retrieval, Quality on Demand, and automated deployment checks.

Future Scope



Scaling to dynamic populations, establishing commercial operator agreements, implementing robust subscription lifecycles, and ensuring regulatory compliance.

Demo Walkthrough and Commercial Value

Leveraging network connectivity for real-time safety and operational efficiency



NOKIA INTEGRATION

Validates live network connectivity and device reporting against technical documentation.



LIVE SIMULATION

Demonstrates real-time crowd management, alarm triggering.



DUAL-DEMO APPROACH

Showcases both network-integrated calmness and incident-response capabilities without requiring operator credentials.



TARGET MARKETS

Designed for stadium operators, city safety authorities, transport hubs, and mobile operators.



ZERO CAPEX

Utilizes existing network reach, eliminating the need for cameras, drones, or maintenance contracts.



SCALABLE COST MODEL

Expenses scale with incidents rather than coverage area, ensuring cost-neutral expansion.



EXTENDED COVERAGE

Provides protection for transit links and peripheral areas without additional hardware.



ACTIVE INTERVENTION

Uses 'Quality on Demand' to dynamically change network behavior for active incident management.

Impact and business value

Performance metrics and operational benefits of CROVIA

Metric	CROVIA, measured	Compared with today
Detection lead time	Plus 8.9 minutes median	Radio call after visible crush
Dangerous crowds caught	33 out of 33	N/A
False alarms	0 out of 117 safe cases	N/A
API cost per evening	480 (quiet) / 1,890 (dangerous)	14,130 (7.5x more expensive)
Headcount accuracy	20% median error	N/A

CROVIA ENABLES PROACTIVE INTERVENTION, SIGNIFICANTLY REDUCES OPERATIONAL COSTS, AND PROVIDES AUDITABILITY THROUGH DETERMINISTIC DATA.

Team & Roles

Placeholder profiles — swap in each teammate's name, role, and focus area

01

**Mohamed
Abdalla**

- **Backend Design**
- **Business Modelling**
- **Interface Development**

02

**Abdelrahman
Ibrahim**

- Developed the core idea of the application
- Designed the engine core and the backend architecture and decisions
- Deployed the app and verified it
- Tested the app across different scenarios to make sure it worked correctly
- Developed the main interface for the user and the authority

03

Ahmed Saied

- **AI Layer Structuring**
- **Database Management**
- **Interface Development**

04

**Nour
Mahmoud**

- **AI Layer Structuring**
- **Business Modelling**
- **Interface Development**

05

**Ahmed
ElMansy**

- **Project Deployment**
- **Business Modelling**
- **Interface Development**

Thank You



Questions and discussion welcome

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